

Vectors in n Dimensions

ANSWER KEY



Vector addition and scalar multiplication

- 1 Let $\mathbf{u} = \langle 2, -1, 3 \rangle$, $\mathbf{v} = \langle 1, 4, -2 \rangle$. Find $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
 $\mathbf{u} + \mathbf{v} = \langle 3, 3, 1 \rangle$. $3\mathbf{u} = \langle 6, -3, 9 \rangle$, $2\mathbf{v} = \langle 2, 8, -4 \rangle$, so $3\mathbf{u} - 2\mathbf{v} = \langle 4, -11, 13 \rangle$.
 - 2 Let $\mathbf{a} = \langle 1, 0, -2, 3 \rangle$, $\mathbf{b} = \langle -1, 2, 1, 0 \rangle$ (vectors in $\mathbb{R}^4$). Find $\mathbf{a} + 2\mathbf{b}$.
 $2\mathbf{b} = \langle -2, 4, 2, 0 \rangle$, so $\mathbf{a} + 2\mathbf{b} = \langle -1, 4, 3, 3 \rangle$.
 - 3 Solve for $\mathbf{x}$: $\mathbf{x} + \langle 2, -3 \rangle = \langle 5, 1 \rangle$.
 $\mathbf{x} = \langle 5, 1 \rangle - \langle 2, -3 \rangle = \langle 3, 4 \rangle$.
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Dot product and norm

- 4 For $\mathbf{u} = \langle 1, 2, 2 \rangle$, $\mathbf{v} = \langle 3, -1, 0 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$ and $|\mathbf{u}|$.
 $\mathbf{u} \cdot \mathbf{v} = 1(3) + 2(-1) + 2(0) = 1$. $|\mathbf{u}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$.
 - 5 Find the angle between $\mathbf{u} = \langle 1, 0, 0 \rangle$ and $\mathbf{v} = \langle 1, 1, 0 \rangle$ using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.
 $\mathbf{u} \cdot \mathbf{v} = 1$, $|\mathbf{u}| = 1$, $|\mathbf{v}| = \sqrt{2}$. $\cos \theta = \frac{1}{\sqrt{2}}$, so $\theta = 45^\circ$ ($\pi/4$).
 - 6 For $\mathbf{u} = \langle 2, 2, -1 \rangle$, $\mathbf{v} = \langle 0, 3, 4 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$, $|\mathbf{u}|$, and $|\mathbf{v}|$.
 $\mathbf{u} \cdot \mathbf{v} = 0 + 6 - 4 = 2$. $|\mathbf{u}| = \sqrt{4 + 4 + 1} = 3$. $|\mathbf{v}| = \sqrt{0 + 9 + 16} = 5$.
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The Cauchy–Schwarz Inequality

- 7 State the Cauchy–Schwarz Inequality.
For all vectors $\mathbf{u}, \mathbf{v}$: $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$.
 - 8 Verify the Cauchy–Schwarz Inequality for $\mathbf{u} = \langle 3, 4 \rangle$, $\mathbf{v} = \langle 1, 2 \rangle$.
 $\mathbf{u} \cdot \mathbf{v} = 3 + 8 = 11$. $|\mathbf{u}| = 5$, $|\mathbf{v}| = \sqrt{5}$, so $|\mathbf{u}||\mathbf{v}| = 5\sqrt{5} \approx 11.18$. Since $11 \leq 11.18$, the inequality holds (it is strict here because $\mathbf{u}$ and $\mathbf{v}$ are not parallel).
 - 9 For which pairs $\mathbf{u}, \mathbf{v}$ does Cauchy–Schwarz hold with EQUALITY? Verify equality for $\mathbf{u} = \langle 2, 4 \rangle$, $\mathbf{v} = \langle 1, 2 \rangle$.
Equality holds exactly when $\mathbf{u}$ and $\mathbf{v}$ are parallel (one is a scalar multiple of the other). Here $\mathbf{u} = 2\mathbf{v}$:
 $\mathbf{u} \cdot \mathbf{v} = 2 + 8 = 10$; $|\mathbf{u}| = \sqrt{20} = 2\sqrt{5}$, $|\mathbf{v}| = \sqrt{5}$, so $|\mathbf{u}||\mathbf{v}| = 2(5) = 10$. Indeed $10 = 10$.
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Unit vectors

- 10 Find the unit vector in the direction of $\mathbf{v} = \langle 3, -4 \rangle$.
 $|\mathbf{v}| = \sqrt{9 + 16} = 5$. Unit vector: $\langle \frac{3}{5}, -\frac{4}{5} \rangle$.
- 11 Find a unit vector in the direction of $\mathbf{v} = \langle 1, 2, 2 \rangle$.
 $|\mathbf{v}| = \sqrt{1 + 4 + 4} = 3$. Unit vector: $\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3} \rangle$.
- 12 Find a vector of length 10 in the same direction as $\mathbf{v} = \langle 3, 4 \rangle$.
 $|\mathbf{v}| = 5$, so the unit vector is $\langle \frac{3}{5}, \frac{4}{5} \rangle$. Scaling to length 10: $10\langle \frac{3}{5}, \frac{4}{5} \rangle = \langle 6, 8 \rangle$.

Orthogonality

- 13 Determine whether $\mathbf{u} = \langle 2, -3, 1 \rangle$ and $\mathbf{v} = \langle 1, 2, 4 \rangle$ are orthogonal.
 $\mathbf{u} \cdot \mathbf{v} = 2(1) + (-3)(2) + 1(4) = 2 - 6 + 4 = 0$. Since the dot product is 0, they are orthogonal.
- 14 For $\mathbf{u} = \langle 1, -1, 2 \rangle$, find $|\mathbf{u}|$ and the unit vector $\hat{\mathbf{u}}$.
 $|\mathbf{u}| = \sqrt{1 + 1 + 4} = \sqrt{6}$. $\hat{\mathbf{u}} = \langle \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \rangle$.
- 15 Find a value of k so that $\mathbf{u} = \langle k, 3 \rangle$ is orthogonal to $\mathbf{v} = \langle 4, -2 \rangle$.
Need $\mathbf{u} \cdot \mathbf{v} = 0$: $4k - 6 = 0 \Rightarrow k = \frac{3}{2}$.

The triangle inequality and vector geometry

- 16 State the Triangle Inequality for vectors.
For all vectors $\mathbf{u}, \mathbf{v}$: $|\mathbf{u} + \mathbf{v}| \leq |\mathbf{u}| + |\mathbf{v}|$.
- 17 Verify the Triangle Inequality for $\mathbf{u} = \langle 3, 0 \rangle$, $\mathbf{v} = \langle 0, 4 \rangle$.
 $\mathbf{u} + \mathbf{v} = \langle 3, 4 \rangle$, so $|\mathbf{u} + \mathbf{v}| = 5$. $|\mathbf{u}| + |\mathbf{v}| = 3 + 4 = 7$. Since $5 \leq 7$, the inequality holds.

Projections

- 18 The projection of $\mathbf{u}$ onto $\mathbf{v}$ is $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \mathbf{v}$. Find $\text{proj}_{\mathbf{v}} \mathbf{u}$ for $\mathbf{u} = \langle 3, 4 \rangle$, $\mathbf{v} = \langle 1, 0 \rangle$.
 $\mathbf{u} \cdot \mathbf{v} = 3$, $|\mathbf{v}|^2 = 1$. $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{3}{1} \langle 1, 0 \rangle = \langle 3, 0 \rangle$.
- 19 Find $\text{proj}_{\mathbf{v}} \mathbf{u}$ for $\mathbf{u} = \langle 4, 3 \rangle$, $\mathbf{v} = \langle 1, 1 \rangle$.
 $\mathbf{u} \cdot \mathbf{v} = 7$, $|\mathbf{v}|^2 = 2$. $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{7}{2} \langle 1, 1 \rangle = \langle \frac{7}{2}, \frac{7}{2} \rangle$.

Linear combinations and span

- 20 Write $\mathbf{w} = \langle 7, 4 \rangle$ as a linear combination $a\mathbf{u} + b\mathbf{v}$ of $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle 0, 1 \rangle$.
Since $\mathbf{u}, \mathbf{v}$ are the standard basis vectors, $\mathbf{w} = 7\mathbf{u} + 4\mathbf{v}$.

- 21 Write $\mathbf{w} = \langle 5, 1 \rangle$ as a linear combination $a\mathbf{u} + b\mathbf{v}$ of $\mathbf{u} = \langle 1, 1 \rangle$ and $\mathbf{v} = \langle 1, -1 \rangle$.
Need $a + b = 5$ and $a - b = 1$. Adding: $2a = 6 \Rightarrow a = 3$; then $b = 2$. So $\mathbf{w} = 3\mathbf{u} + 2\mathbf{v}$.
- 22 Explain briefly why every vector in $\mathbb{R}^2$ can be written as a linear combination of $\mathbf{u} = \langle 1, 1 \rangle$ and $\mathbf{v} = \langle 1, -1 \rangle$ (i.e., why they span $\mathbb{R}^2$).
 $\mathbf{u}$ and $\mathbf{v}$ are not parallel (neither is a scalar multiple of the other), so they point in two genuinely different directions in the plane. Two non-parallel vectors in $\mathbb{R}^2$ always span the whole plane, since any target vector's components along each direction can always be solved for (as in the two questions above, the resulting 2×2 system always has a unique solution).
- 23 Is $\mathbf{w} = \langle 3, 6 \rangle$ in the span of $\mathbf{u} = \langle 1, 2 \rangle$ alone? Justify your answer.
Yes: $\mathbf{w} = 3\mathbf{u}$, since $3\langle 1, 2 \rangle = \langle 3, 6 \rangle$. A single nonzero vector's span is the entire line through the origin in its direction, and $\mathbf{w}$ lies on that line.
- 24 Explain why $\mathbf{u} = \langle 1, 2 \rangle$ and $\mathbf{v} = \langle 2, 4 \rangle$ do NOT span all of $\mathbb{R}^2$.
 $\mathbf{v} = 2\mathbf{u}$, so $\mathbf{u}$ and $\mathbf{v}$ are parallel — every linear combination $a\mathbf{u} + b\mathbf{v} = (a + 2b)\mathbf{u}$ is just a scalar multiple of $\mathbf{u}$, so their span is only the line through the origin in the direction of $\mathbf{u}$, not the whole plane.
- 25 Determine whether $\mathbf{w} = \langle 1, 1, 1 \rangle$ is a linear combination of $\mathbf{u} = \langle 1, 0, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 0 \rangle$.
No. Any combination $a\mathbf{u} + b\mathbf{v} = \langle a, b, 0 \rangle$ always has a 0 in the third coordinate, but $\mathbf{w}$ has a 1 there — so $\mathbf{w}$ is not in the span of $\mathbf{u}, \mathbf{v}$.
- 26 Write $\mathbf{w} = \langle 4, -1, 7 \rangle$ as a linear combination of $\mathbf{u} = \langle 1, 0, 0 \rangle$, $\mathbf{v} = \langle 0, 1, 0 \rangle$, and $\mathbf{z} = \langle 0, 0, 1 \rangle$, and explain why this is always possible for ANY vector in $\mathbb{R}^3$.
 $\mathbf{w} = 4\mathbf{u} - 1\mathbf{v} + 7\mathbf{z}$. This is always possible for any vector in $\mathbb{R}^3$ because $\mathbf{u}, \mathbf{v}, \mathbf{z}$ are the standard basis vectors: every vector's own coordinates ARE exactly its coefficients in this combination, so $\{\mathbf{u}, \mathbf{v}, \mathbf{z}\}$ spans all of $\mathbb{R}^3$ by definition.
- 27 How many vectors, at minimum, are needed to span $\mathbb{R}^n$? Justify briefly.
At least n vectors are needed. Fewer than n vectors can span at most a subspace of dimension less than n (a proper subset of $\mathbb{R}^n$), so they cannot reach every vector in $\mathbb{R}^n$; the standard basis shows n vectors suffice.
- 28 Determine whether $\mathbf{u} = \langle 1, 2, -1 \rangle$, $\mathbf{v} = \langle 2, 1, 1 \rangle$, and $\mathbf{w} = \langle 3, 3, 0 \rangle$ span $\mathbb{R}^3$, given that $\mathbf{w} = \mathbf{u} + \mathbf{v}$.
No. Since $\mathbf{w} = \mathbf{u} + \mathbf{v}$, $\mathbf{w}$ contributes no new direction beyond $\mathbf{u}$ and $\mathbf{v}$ — every linear combination $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = a\mathbf{u} + b\mathbf{v} + c(\mathbf{u} + \mathbf{v}) = (a + c)\mathbf{u} + (b + c)\mathbf{v}$ is really just a combination of $\mathbf{u}$ and $\mathbf{v}$ alone. Two vectors can span at most a 2-dimensional subspace (a plane through the origin) of $\mathbb{R}^3$, not the full three-dimensional space.
- 29 Give an example of three vectors in $\mathbb{R}^3$ that DO span $\mathbb{R}^3$, and briefly justify why they work.
The standard basis $\mathbf{e}_1 = \langle 1, 0, 0 \rangle$, $\mathbf{e}_2 = \langle 0, 1, 0 \rangle$, $\mathbf{e}_3 = \langle 0, 0, 1 \rangle$ works: any vector $\langle a, b, c \rangle$ equals $a\mathbf{e}_1 + b\mathbf{e}_2 + c\mathbf{e}_3$ directly, so every vector in $\mathbb{R}^3$ is reachable.

- 30** Summarize in one sentence the difference between a set of vectors being linearly independent and a set of vectors spanning a space.

Spanning is about reaching every vector in the space (enough vectors, covering all directions); linear independence is about having no redundant vectors (none is a combination of the others) — a basis needs both properties at once.